SAMPLE

Earth's Dynamic Systems: The Oceans

A Biography of the World Ocean

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Introduction

Stand on any shore and the ocean looks like one thing: a single flat blue horizon-to-horizon body. It is the oldest illusion in geography. The world ocean is a system of interacting basins with its own geology, its own weather, its own deserts and rainforests, its own mountain chains longer than any on land, and its own dark interior that remains less mapped, by resolution, than the surface of the Moon. It holds ninety-seven percent of the free water on the planet, covers seventy-one percent of its surface, and takes up more than ninety-nine percent of the volume in which life can live. Almost every striking fact about the Earth, from the mildness of its climate to the oxygen in its air, routes through the sea.

This book is about the ocean as a whole object: where its water came from, how its basins were built and how they keep being rebuilt, what its water is made of and how its layers stack, what lives in it from the sunlit skin to the hadal trenches, and how a species that lived for most of its history afraid of the deep came to explore it with sounding lines, submersibles, satellites, and robot floats. It is a biography of the largest feature of the planet, written from the evidence.

The story divides naturally into movements. The first is historical: how the ocean, long a boundary to be feared or a road to be crossed, became an object of science, from the sounding lines of the nineteenth century to the great mid-century expeditions that revealed the seafloor as geology in motion. The second is physical and chemical: the anatomy of the water column, the salt and the heat it carries, and the layered structure of light and life that makes the ocean, in effect, a stack of very different worlds at different depths. The third is biological: the sunlit pastures, the vast twilight migrations, and the alien ecosystems of hot vents and whale falls that live on chemical energy with no help from the sun. The fourth is the ocean's role as the planet's flywheel: how it stores heat, trades water with the sky, and takes up the carbon dioxide that human industry pours into the air, and what those services cost as the bills come due.

A route map may help before the road begins. Chapter 1 follows the sea's discovery as a single object; Chapters 2 and 3 build its anatomy and its working characteristics; Chapter 4 states the theories that hold the picture together and Chapter 5 the instruments that measure it; Chapter 6 gathers the features that organize the rest, and Chapter 7 carries the story to the ocean's frozen ends; Chapter 8 sets the sea in its larger contexts; Chapters 9 and 10 record the present state of the research and the discoveries it has made; Chapter 11 turns from space to time and reads the ocean's own past from the rock; Chapters 12 and 13 face the open problems and the near future; Chapter 14 keeps the cabinet of wonders; and Chapter 15 states the significance. The reader who wants one thread can follow the water: it arrives, it circulates, it freezes at the ends, it records itself, and it is called to account.

A note on scope is owed at the start. Currents have their own companion volume in this series, as do tides, and the reader who wants the engine of ocean circulation in detail should go there; here the moving water appears as one thread among many. The plan of this book runs straight down: it begins at the surface, descends through the zones to the deepest trench, and ends with what the ocean's condition means for the century now underway.

One habit of ocean science deserves to be stated plainly, because the chapters lean on it repeatedly: almost everything known about the deep water has been learned indirectly, by inference from instruments, in the last seventy years. For all of history before that, and for most of history after, the ocean below the reach of light was a matter of speculation, and honest oceanographers are quick to say that the map is still mostly blank. The blankness is not a failure. It is the standing invitation the sea extends, and the last chapters record how it is being taken up.

Chapter 1 begins with the water itself: where nearly a billion and a half cubic kilometers of it came from, and how the planet arranged to keep it.

Chapter 1: The Discovery of the Oceans

Before the ocean could be studied, it had to be noticed as a single thing, and that noticing took most of recorded history.

The ancient world treated the sea as a boundary and a hazard. The Mediterranean, whose very name means the sea in the middle of lands, was a highway, but the ocean beyond the Pillars of Hercules was the place where maps stopped, the water of myth rather than of measurement. Ptolemy's geography, which governed the Western picture of the world for over a thousand years, shrank the unknown ocean into a modest band around a land-dominated

globe. When the age of sail began in the fifteenth century, ocean crossing was a feat of navigation rather than of understanding: the Portuguese and Spanish caravels crossed the Atlantic routinely while possessing almost no idea how deep it was, what its floor looked like, or why its water was salt.

The birth of oceanography as a data practice came from an unexpected quarter: the needs of ships and trade. Matthew Fontaine Maury, an American naval officer whose career ashore began after an injury ended his sea duty, assembled the logbooks of hundreds of voyages into the first systematic charts of winds and sailing routes in the 1840s and 1850s, publishing them in a work that made him, in the common phrase, the father of oceanography. His insight was the founding method of the discipline: the ocean is too large for one observer, so knowledge must be pooled, standardized, and mapped. The method mattered more than any single chart.

The telegraph gave the measurement of the deep its first commercial contract. The plan to lay a working cable across the Atlantic in the 1850s required a route, and a route required soundings, so the navies of Britain and the United States ran lines across the North Atlantic in the years before the first cable of 1858, work that produced the discovery of the telegraph plateau, a comparatively shallow band across the middle of the ocean, and the first suspicion, decades ahead of its proof, that the heart of the sea was not a flat abyss. The cable of 1866 succeeded where the earlier attempts had failed, and the soundings taken for its sake are among the ancestors of the modern seafloor map.

The scientific ocean came into existence with a single expedition, the voyage of HMS Challenger between 1872 and 1876. A converted British warship, refitted with laboratories and winches, spent three and a half years circling the globe, dredging the floor, sampling the water, and measuring depth by sounding line. The cruise covered nearly seventy thousand nautical miles, took hundreds of depth soundings and water samples at every ocean, and returned with crates of specimens that took twenty years and dozens of scientists to describe; the resulting report ran to fifty volumes. Two Challenger soundings still anchor the science: the 1875 measurement, near the Mariana Islands in the Pacific, of a depth of over eight thousand meters, and the enormous haul of life from every depth, which destroyed the older belief that the deep sea was lifeless. The deepest known point of the ocean still carries the ship's name.

A third discovery, quieter than the expeditions but equally decisive, concerned the water itself. Analysis of seawater, refined through the nineteenth century by William Dittmar on the Challenger samples and others, established the fact that makes modern ocean chemistry possible: seawater everywhere, from the Arctic to the tropics, contains the same principal salts in almost exactly the same proportions. The total amount of salt varies slightly, from about thirty-three to thirty-seven grams per kilogram, but the recipe barely changes. This constancy, because the ocean mixes thoroughly on long timescales, lets a single measurement of one constituent, usually chlorinity, determine the salinity of the whole sample, and it testifies that the ocean behaves as one connected chemical system, not a set of separate lakes.

The twentieth century added the geophysical ocean. Echo sounding, introduced widely in the 1920s, replaced the laborious sounding line with acoustics, and the German Meteor expedition of 1925 to 1927 produced the first continuous profiles of an entire ocean basin, revealing a floor far more rugged than anyone had pictured. In the decades after the Second World War, sonar surveys, cores, and magnetometers assembled the picture that underlies the modern science: the ocean basins are the youngest surfaces on Earth, spreading from submerged ridges, sinking at trenches, and rebuilt on a timescale of about two hundred million years. The discovery that the seafloor itself is mobile is told in full in the chapter on the science behind the ocean, because it changed everything, including the meaning of the water above.

The result of a century and a half of such work is a picture with three fixed points worth stating at the outset. The ocean holds about 1.335 billion cubic kilometers of water, which is about ninety-seven percent of all the free water on the planet. It covers about seventy-one percent of the surface, with an average depth of about 3,700 meters. And it is, by volume, more than ninety-nine percent of the living space on Earth: every forest, field, and city stands in the thin remainder above its surface. The discovery of the oceans, in the end, is the discovery that the solid planet we inhabit is a crust over a water world's statistics.